

Dr. Hao-Chun (Chad) Yang

SENIOR AI/ML ENGINEER

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Professional Experience

Aptiv

Senior Machine Learning Engineer

Germany
July 2024 - Present

- Led a five-engineer team developing an end-to-end multimodal sensor-fusion perception system for Level 4 autonomous driving.
- Architected and optimized Transformer-based 3D perception modules with TensorRT and ONNX for edge deployment, improving performance by more than 8% over production benchmarks.
- Partnered with MLOps, data, and KPI teams to integrate Airflow and MLflow pipelines supporting robust data quality and efficient model training.

University of Tübingen / Donders Institute

Postdoctoral Researcher, Machine Learning

Germany / Netherlands
Mar 2023 - Jul 2024

- Researched self-supervised and unsupervised representation learning for long-tailed neuroimaging disease detection.
- Improved psychosis detection by 6.7% AUROC using a self-supervised masked mesh vision transformer.
- Investigated vision-language foundation-model fine-tuning for large-scale neurodegenerative disease detection.

C-Media Electronics, Inc.

Senior Machine Learning Scientist

Taipei, Taiwan
Jan 2020 - Dec 2021

- Led a team of five developing a real-time fully convolutional speech dereverberation engine, achieving an SRMR score of 4.774.
- Oversaw the full R&D pipeline, including data collection and synthesis, model and metric design, error analysis, and model compression.
- Pioneered speech cloning from previously unseen sources using generative adversarial networks.

Inventec AI Center

Machine Learning Scientist, Cardiovascular Disease Detection

Taipei, Taiwan
Jan 2021 - Dec 2021

- Led development of an end-to-end heart-disease anomaly detection system using time-series, multi-lead ECG.
- Proposed Mixed-Domain Self-Attention ResNet (MDARsn) to handle missing and anomalous ECG data for robust multilabel detection.
- Received the Best Challenge Poster Award in the PhysioNet/CinC Challenge 2021 and secured US patent US20230153575A1.

Institute for Information Industry (III)

Lead ML Consultant

Taipei, Taiwan
Mar 2020 - Nov 2020

- Led a team of five developing a deep video-retrieval system to accelerate fake-news screening.
- Proposed a video-retrieval system with 95.1% retrieval precision and orchestrated its deployment with Docker.
- Enabled deployment at Taiwan FactCheck Center and MyGoPen, two leading nonprofit fact-checking organizations.

National Health Research Institutes

Senior Machine Learning Scientist, Domain Transfer and Privacy

Taiwan
Sep 2020 - Apr 2022

- Introduced a privacy-aware federated learning approach (FedCM) to improve Alzheimer disease detection across multiple sources.
- Identified and differentiated key regions of interest related to normal brain degeneration and Alzheimer disease.
- Mentored students on robust multi-site ADHD detection using a conditional-adversarial domain adaptation network (CDAN).

Machine Learning Scientist, Neurocognitive Assessment Using Brain MRI

Sep 2018 - Sep 2020

- Spearheaded an MRI research pipeline spanning preprocessing with FreeSurfer and SPM, modeling, and explainable AI with Grad-CAM++.
- Increased three-class Alzheimer disease detection by 8.98% UAR using a 3D autoencoder with multi-view contrastive loss (BSEN).
- Achieved state-of-the-art 80.2% UAR in binary face-processing ability classification using event-contrastive Node2Vec embeddings.

Behavioral Informatics & Interaction Computation Lab (BIIC)

Doctoral Researcher, Affective Electrophysiology Responses under Audio / Visual Stimuli

Taiwan
Sep 2018 - Apr 2022

- Enhanced physiological emotion recognition by 3.5% UAR through Transformer-based personalized modeling.
- Achieved an 8% UAR increase in personality recognition using a multi-head attention graph convolutional network.
- Studied subjective and intended emotion responses from ECG, EDA, and EEG signals with Shapley analysis.

Education

National Tsing Hua University (NTHU)

Ph.D., Electrical Engineering

Hsinchu, Taiwan

Sep 2016 - Apr 2022

National Tsing Hua University (NTHU)

B.S., Electrical Engineering

Hsinchu, Taiwan

Sep 2012 - Jun 2016

Skills

Programming: Python, **CUDA**, C++, Bash, SQL, Matlab, YAML

DevOps | Cloud: Git, Docker, CI/CD | **Azure**, AWS **MLOps | Distributed Training:** MLflow, Airflow, W&B, Databricks

Deep Learning | Optimization: Pytorch (Pytorch Geometric, Pytorch3D, Lightning), TensorRT, Tensorflow, ONNX, Huggingface | Optuna

Framework: Transformer, Graph Neural Network (GNN), Large Language Models (LLMs), Variational Autoencoder (VAE), Recurrent Neural Network (RNN), Diffusion Models, Generative Adversarial Network (GAN)

Data Science: Scikit-Learn, Xgboost, OpenCV, Numpy, Pandas, Matplotlib, SHAP

Professional Interest: Self-supervised Learning, Generative Models, Domain Adaptation, Anomaly Detection

Teaching

National Tsing Hua University

Teaching Assistant

- 10720IMS503100: Artificial Intelligence and Entrepreneurship
- 10710EE366200: Digital Signal Processing Laboratory
- 10620EE306001: Probability
- 10610EE648500: Computer Vision

Honors & Awards

2022	Program Committee Member , IEEE ICKII, IEEE ICEIB, IARIA CENTRIC	
2021	Best Challenge Poster , PhysioNet/CinC Challenge	Computing in Cardiology Society
2019	Travel Grant , 2019 ICASSP SPS Travel Grant	IEEE Signal Processing Society
2018	Scholarship , Paper Presentation Scholarship	ACLCLP, Taiwan
2018	Scholarship , AI Scholarship	Adbertech Inc., Taiwan
2018	Scholarship , President Scholarship	National Tsing Hua University
2015	Scholarship , Summer Academic Exchange Program	Fudan University, China

Selected Publications ([Google Scholar Profile](#))

JOURNAL

- [1] A Media-Guided Attentive Graphical Network for Personality Recognition Using Physiology
Hao-Chun Yang and Chi-Chun Lee
IEEE Transactions on Affective Computing (2021) pp. 1–12021

PEER-REVIEWED CONFERENCE/WORKSHOP PAPER

- [1] **Hao-Chun Yang**, Thomas Wolfers, Ole Andreassen, Lars Tjelta Westlye, A.F. Marquand, and C.F. Beckmann, “Learning Cortical Anomaly through Masked Encoding for Unsupervised Heterogeneity Mapping”
2024 IEEE International Symposium on Biomedical Imaging (ISBI)
- [2] Ya-Lin Huang, **Hao-Chun Yang**, and Chi-Chun Lee, “Federated Learning via Conditioned Mutual Learning for Alzheimer Disease Classification on T1w MRI”
43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society, EMBC 2021, (Virtual), Nov 1-5, 2021
- [3] **Hao-Chun Yang**, Wan-Ting Hsieh, and Pei-Chun Chen, “A Mixed-Domain Self-Attention Network for Multilabel Cardiac Irregularity Classification Using Reduced-Lead Electrocardiogram”
Computing in Cardiology, CinC 2021, Brno, Czech Republic, September 12-15
- [4] Woan-Shiuan Chien, **Hao-Chun Yang**, and Chi-Chun Lee, “Cross Corpus Physiological-based Emotion Recognition Using a Learnable Visual Semantic Graph Convolutional Network”
MM '20: The 28th ACM International Conference on Multimedia, ACM MM 2020, Virtual Event / Seattle, WA, USA, October 12-16, 2020
- [5] Wan-Ting Hsieh, Jeremy Lefort-Besnard, **Hao-Chun Yang**, Li-Wei Kuo, and Chi-Chun Lee, “Behavior Score-Embedded Brain Encoder Network for Improved Classification of Alzheimer Disease Using Resting State fMRI”
42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society, EMBC 2020, Montreal, QC, Canada, July 20-24, 2020
- [6] Ya-Lin Huang, Wan-Ting Hsieh, **Hao-Chun Yang**, and Chi-Chun Lee, “Conditional Domain Adversarial Transfer for Robust Cross-Site ADHD Classification Using Functional MRI”
2020 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2020, Barcelona, Spain, May 4-8, 2020
- [7] **Hao-Chun Yang** and Chi-Chun Lee, “A Siamese Content-Attentive Graph Convolutional Network for Personality Recognition Using Physiology”
2020 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2020, Barcelona, Spain, May 4-8, 2020
- [8] Wan-Ting Hsieh, **Hao-Chun Yang**, Fu-Sheng Tsai, Chon-Wen Shyi, and Chi-Chun Lee, “An Event-contrastive Connectome Network for Automatic Assessment of Individual Face Processing and Memory Ability”
IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2019, Brighton, United Kingdom, May 12-17, 2019
- [9] **Hao-Chun Yang** and Chi-Chun Lee, “An Attribute-invariant Variational Learning for Emotion Recognition Using Physiology”
IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2019, Brighton, United Kingdom, May 12-17, 2019
- [10] **Hao-Chun Yang** and Chi-Chun Lee, “Annotation Matters: A Comprehensive Study on Recognizing Intended, Self-reported, and Observed Emotion Labels using Physiology”
8th International Conference on Affective Computing and Intelligent Interaction, ACII 2019, Cambridge, United Kingdom, September 3-6, 2019
- [11] Wan-Ting Hsieh, **Hao-Chun Yang**, Ya-Tse Wu, Fu-Sheng Tsai, Li-Wei Kuo, and Chi-Chun Lee, “Integrating Perceivers Neural-Perceptual Responses Using a Deep Voting Fusion Network for Automatic Vocal Emotion Decoding”
2018 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2018, Calgary, AB, Canada, April 15-20, 2018
- [12] Yi-Ying Kao, Hsiang-Ping Hsu, Chien-Feng Liao, Yu Tsao, **Hao-Chun Yang**, Jeng-Lin Li, Chi-Chun Lee, Hung-Shin Lee, and Hsin-Min Wang, “Automatic Detection of Speech Under Cold Using Discriminative Autoencoders and Strength Modeling with Multiple Sub-Dictionary Generation”
16th International Workshop on Acoustic Signal Enhancement, IWAENC 2018, Tokyo, Japan, September 17-20, 2018
- [13] Fu-Sheng Tsai, **Hao-Chun Yang**, Wei-Wen Chang, and Chi-Chun Lee, “Automatic Assessment of Individual Culture Attribute of Power Distance Using a Social Context-Enhanced Prosodic Network Representation”
Interspeech 2018, 19th Annual Conference of the International Speech Communication Association, Hyderabad, India, 2-6 September 2018
- [14] **Hao-Chun Yang**, Fu-Sheng Tsai, Yi-Ming Weng, Chip-Jin Ng, and Chi-Chun Lee, “A Triplet-Loss Embedded Deep Regressor Network for Estimating Blood Pressure Changes Using Prosodic Features”
2018 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2018, Calgary, AB, Canada, April 15-20, 2018